

ASSEMBLYAI VOICE AGENT HACKATHON

The warning has to land.

A voice agent that phones outdoor workers when heat turns dangerous, then records whether they understood.

`heat-warning-agent.vercel.app`

`github.com/ubaidai/heat-warning-agent`

The forecast existed. It never reached the scaffold.

Karachi's 2015 heatwave killed more than a thousand people in a single week, most of them working outdoors.

Heat is the one disaster we can see coming days ahead. The failure is never prediction. It is delivery.



1,000+

Dead in one week, June 2015.
The heat was forecast.

A worker who says yes has not understood

Agreeing with an authoritative voice on a phone is the ordinary human response, not the exception.

A naive flow files that yes as a delivered warning. The employer ends up holding a record full of agreement that meant nothing, **which is worse than no record at all.**

CALL

connectedEND CALL

agent Imran, this is an automated safety call from Al-Habib Construction. The heat at your site has passed a dangerous level and you need to stop work now. Tell me what you are going to do.

worker Yes.

Understood

NOT ESTABLISHED

"Yes."

Stopped work

NOT ESTABLISHED

Water and shade

NOT ESTABLISHED

ONE REAL CALL, FOUR TURNS

So it never asks a question that yes can answer

AGENT The heat has passed a dangerous level. Tell me what you are going to do.

WORKER Yes.
`record_understanding   established: false   evidence: "Yes."`

AGENT I need more than a yes. Tell me what you are going to do in the next few minutes.

WORKER I am coming down off the scaffold. I will sit in the shade by the gate.
`record_understanding   established: true`

What a call has to establish

Understood the warning

Only when the worker says something showing comprehension. Their own words are kept beside the judgement, so a reviewer can check the call rather than trust it.

Stopped work, or why not

"I am paid per floor" is the most useful thing a call can learn, and no thermometer reports it. A refusal is its own outcome, never a partial success.

Water and shade, both

A bottle in your hand with nowhere out of the sun is not a safe worker, so one without the other does not count.

Distress, which stops everything

Dizziness, nausea or confusion abandons the script on the spot and hands the call to a human.

Nothing is assumed

Every fact starts `null`, meaning not established, which is a different state from false.

A call that drops before the water question must never record that the worker had no water. Water and shade are carried separately, so the record says **which half is missing**.

CALL			connected	END CALL
agent Imran, this is an automated safety call from Al-Habib Construction. The heat at your site has passed a dangerous level and you need to stop work now. Tell me what you are going to do.				
worker Yes.				
agent I need to know you understand the danger, Imran. Do not just say yes. Tell me what you are going to do in the next few minutes.				
worker I am coming down off the scaffold. I will sit in the shade by the gate.				
Understood ESTABLISHED "I am coming down off the scaffold. I will sit in the shade by the gate."	Stopped work ESTABLISHED "I am coming down off the scaffold."	Water and shade WATER NOT ESTABLISHED, SHADE YES "I will sit in the shade by the gate."		

What the employer is left holding

An alert that was sent proves nothing. A call where someone said what they were going to do, in their own words, with the outcome attached, is what an employer produces when a regulator asks whether they acted.

```
{
  "status":      "refused_or_unable_to_stop",
  "reached":     true,
  "factsEstablished": 2,
  "of":          3,
  "blocker":     "Paid per floor",
  "distress":    null,
  "durationSeconds": 74
}
```

Built on the AssemblyAI Voice Agent API

One WebSocket carries everything: audio both ways, transcripts, and tool calls. PCM16 at 24 kHz, four function tools writing the record as the call runs.

No separate speech-to-text, model and text-to-speech to stitch together, which is the reason to use this API rather than assembling the stack by hand.

1

SOCKET, NOT THREE
SERVICES

4

TOOLS WRITING THE
RECORD

3

LANGUAGES, EN / HI /
AR

23

TESTS PASSING

What is not done yet

Urdu is served by the Hindi model

AssemblyAI's streaming models do not include Urdu. Spoken Urdu and Hindi diverge in script and formal register rather than in "I have stopped, I have water", but it is an approximation and it needs testing against real Karachi speech before anyone relies on it.

Telephony is not wired up

Calls run in the browser today. The session layer is written against a telephony leg so the swap is one adapter, not a rewrite.

Nobody has been called for real

Every transcript shown here is a live agent session with a simulated worker. Field testing with an employer is the next thing, not a later thing.

TRY IT, IT TAKES A MINUTE

Say yes without meaning it.

Then watch what the record refuses to claim on your behalf.

`heat-warning-agent.vercel.app`

`github.com/ubaidai/heat-warning-agent`

`MIT licensed`